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A.2

1. $\frac{\partial^2 u}{\partial t^2} - \frac{c^2}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) = \frac{\partial^2(ur)}{\partial t^2} - c^2\frac{\partial^2(ru)}{\partial r^2} = 0$

令 $ur = A$, 由波动方程得:

通解:

$ur = A = f(r-ct) + g(r+ct)$

$\Rightarrow u = \frac{1}{r}[f(r-ct) + g(r+ct)]$

A.3

1.

由题意, $\begin{cases} \frac{\partial u}{\partial t} - K\frac{\partial^2 u}{\partial x^2} = 0 & ① \\ u|_{x=0} = 0,\ u|_{x=L} = 0 & ② \\ u|_{t=0} = ax(L-x) & ③ \end{cases}$

令 $u = X(x)T(t)$, 代入①, ②

$\begin{cases} XT' - KX''T = 0 \Rightarrow \frac{X}{X''} = K\frac{T'}{T} = -\lambda & ④ \\ X_{(0)} = 0,\ X_{(L)} = 0 & ⑤ \end{cases}$

由④得, $\begin{cases} X + \lambda X'' = 0 \\ X_{(0)} = 0,\ X_{(L)} = 0 \end{cases}$, $T + \frac{K}{\lambda}T' = 0$ ⑥

$\lambda = 0$ 时, $X \equiv 0$ 非特征值

入≠0时,

$$X = A\sin\sqrt{\lambda}x + B\cos\sqrt{\lambda}x$$

由边界条件 $B=0$, $A\sin\sqrt{\lambda}l=0$

$$\Rightarrow \begin{cases} \lambda_n = \left(\frac{n\pi}{l}\right)^2, n=1,2,\cdots \\ X = \sin\frac{n\pi}{l}x \end{cases}$$

回代入⑥,

$$T + k\left(\frac{n\pi}{l}\right)^2 T = 0$$

解得:

$$T = Ce^{-k\left(\frac{n\pi}{l}\right)^2 t}$$

故特解:

$$u_n(x,t) = Ce^{-k\left(\frac{n\pi}{l}\right)^2 t}\sin\frac{n\pi}{l}x$$

代入初始状态:

$$\sum_{n=1}^{\infty} C\sin\frac{n\pi}{l}x = ax(l-x)$$

同乘 $\sin\frac{m\pi}{l}x$, 积分,

$$\int_0^l ax(l-x)\sin\frac{m\pi}{l}x\,dx = \int_0^l \sum_{n=1}^{\infty} C\sin\frac{n\pi}{l}x\sin\frac{m\pi}{l}x\,dx$$

$$= C\sum_{n=1}^{\infty}\int_0^l \sin\frac{n\pi}{l}x\sin\frac{m\pi}{l}x\,dx$$

$$= \frac{l}{2}\cdot C$$

$$\Rightarrow C = \frac{2}{l}\int_0^l ax(l-x)\sin\frac{n\pi}{l}x\,dx$$

综上, $u_n(x,t) = \frac{2}{l}e^{-k\left(\frac{n\pi}{l}\right)^2 t}\sin\frac{n\pi}{l}x\cdot\int_0^l ax(l-x)\sin\frac{n\pi}{l}x\,dx$